

SUJEN KANCHERLA

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Education

- **University of California San Diego** (Sep 2021 – June 2024): Bachelor of **Data Science** — **GPA: 3.87**.
- **University of California San Diego** (June 2024 – December 2025): Masters in **Machine Learning and Data Science** - Electrical and Computer Engineering — **GPA: 3.33**

Work Experience

AMD

January 2025 – current

Machine Learning Intern

- Used (Generative AI)/ML to predict bugs in post silicon
- working with x86 architecture to enhance ML performance using cpu instruction set architecture

UCSD ARC Lab

July 2024 – March 2025

Graduate Student Researcher

- Working with Nvidia Isaac Sim, in python
- Automating robotic surgery through Reinforcement Learning in simulation

UCSD McAuley Lab

Oct 2023 – April 2024

Student Research Assistant

Remote

- Co-author of paper: *Avoiding Decision Fatigue with AI-Assisted Decision-Making*
- Introduced Decision-based reinforcement learning on sequential data for recommender systems. Used a **GPT-2 transformer** with **Reinforcement Agent**
- Performed Exploratory Data Analysis on sequential decision making data using **pandas** and **seaborn**
- Created baseline model using **Deep-Q learning**, designed the reward strategy, environment and loss function to achieve an accuracy of **80%**

Omnisync INC

Jan 2023 – June 2023

Data Science Intern

Part-time Remote

- leveraged **REST APIs** to extract data into python scripts, employing **AWS** distributed systems (ec2) to significantly reduce processing time
- Performed data wrangling to clean and format data, created schemas and relationships in data to correctly store in both **mySQL** and **weaviate** databases
- Used pretrained Sentence Transformer model and finetuned it using **PyTorch** to implement a semantic search, using vector embeddings stored in the databases. The backend app query search using semantics increased user accessibility and decreased **60%** of average user time on the search system

Intel

June 2022 – Sep 2022

Data Engineering Intern

Full-Time Hybrid

- Used shell scripting in **bash** to add and change legacy scripts that performed graphics benchmarking tests and monitoring graphics card temperature and other metrics
- Developed python package to automate the benchmarking and data generation of graphics metrics during a variety of **KPIs**
- Used **xlsxwriter** to perform data analysis on graphics card temperature during test run of KPI's as a part of our power and performance testing

Projects

Image Resolution Enhancement | [Source Code](#)

Python | PyTorch | GAN | Convolution

- Created **ESRGAN** architecture in PyTorch using the research paper
- Collected and preprocessed image data from div2k dataset for lower and corresponding higher resolution target images.
- Used discriminative and generative **loss functions** to maximize the quality of the output image
- Tuned hyperparameters iteratively with graphing and logging over validation data.
- Created **API** using flask to represent **ONNX** model for faster inference time.

Chess and Tic Tac Toe AI | [Source Code](#)

Python | PyGame | Graph Theory

- Represented game states in Tic Tac Toe and Chess as a **n-ary tree** data structure.
- Implemented a **Depth-limited minimax** algorithm to effectively search for an optimal move at the current state, offering various skill levels of AI
- Created an interactive UI for user to play the games, using **PyGame**

Publications

- **Avoiding Decision Fatigue with AI-Assisted Decision-Making**
Co-author. *Proceedings of the ACM on Human-Computer Interaction*, 2024.
<https://dl.acm.org/doi/10.1145/3627043.3659569>.

Technical Skills

Python, R, SQL, Java, Node.js, HTML, CSS, JavaScript, Linux Bash, Regression, Decision Trees, SVM, Neural Networks, Ensemble Methods, k-NN, Naive Bayes, Decision Forests, Scikit-learn, CNNs, RNNs, TensorFlow, Keras, PyTorch, Hypothesis Testing, Probability, Data Mining, Statistical Testing, Regression Analysis, Tableau, Matplotlib, Seaborn, ggplot2, D3.js, Spark, Dask, MySQL, PostgreSQL, Git, Jupyter Notebook, Visual Studio Code, Azure, AWS Lambda, ECR, EC2